

AYUSH KUMAR SAHU

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Personal Summary

Research-focused undergraduate with expertise in machine learning, computer vision, and signal processing. Leading research on multimodal emotion recognition & lightweight transformer models for edge deployment. Experienced in hardware, robotics & autonomous systems with demonstrated innovation & strong analytical skills.

Education

National Institute of Technology, Rourkela

November 2022 – Present

Bachelor of Technology in Electrical Engineering (CGPA - 8.75 - Rank 7/124)

Rourkela, Odisha

Doon International School, Bhubaneswar

May 2022

Higher Secondary Education: AISCCE – CBSE, Science (PCM) (Percentage - 94.8%)

Bhubaneswar, Odisha

Relevant Coursework

- Machine Learning
- Statistics & Linear Algebra
- Digital Image Processing
- Digital Signal Processing
- Digital Electronics
- Computer Architecture

Work Experience

SMSS Laboratory, IIT Kanpur

May 2024 – July 2024

Robotics Intern - Self Navigating Autonomous Rover (Sponsored by Powergrid Corp. of India)

Kanpur, Uttar Pradesh

- Built the system interface involving RMCS-2303 controllers interfaced to Arduino-uno
- Using UART protocol & PySerial library established serial communication with Arduino & the RPI-4
- Created ROS2 architecture for the RPI-4 to communicate with LIDAR & system interface
- Using the LIDAR data generated a map and enabled self-navigation using SLAM and Nav2.

Projects

Advanced Emotion Recognition Research for edge devices (FER & Multimodal)

Aug 2025 – Present

PyTorch, Transformers, Domain Adaptation, Multimodal Fusion

Research Project

- Engineered a novel **Multi-Head Attention CNN** with **Cross-Contrastive Mappings** to minimize intra-class variance and achieve robust cross-domain alignment and SOTA performances on FER2013, RAF-DB and AffectNet.
- Expanded research scope to **Multimodal Emotion Recognition (MER)**, investigating Transformer-based fusion strategies to align asynchronous Audio-Visual-Text signals.
- Developing a **Lightweight Bottleneck Transformer** with **adaptive token pruning**, significantly reducing computational load for real-time edge inference.
- The work is in advanced research phase for publication and development of a working prototype.

Obstacle Detection and Monocular Depth Estimation

Oct 2025 – Nov 2025

YOLOv8, Depth-Pro, PyTorch, Computer Vision

Project

- Developed an obstacle detection system using a single RGB camera without relying on depth sensors or stereo vision.
- Integrated **YOLOv8** for real-time obstacle localization and **Apple's Depth-Pro for monocular depth estimation** to approximate object distances.
- Trained & evaluated the pipeline using the KITTI dataset, a benchmark dataset widely used in autonomous driving.
- Designed the detection–depth fusion workflow suitable for robotics navigation and ADAS systems in EVs.

Automatic Washing Machine Controller

Jan 2025

Verilog - HDL, Xilinx Vivado

Project

- Implemented a **Finite State Machine (FSM)** to model washing machine operations using **Verilog**.
- Developed and tested design with **testbenches** and simulated functionality on **Xilinx Vivado 2025.1**.
- Prepared design for FPGA implementation, enabling real-time demonstration of automated washing machine control.

Achievements/Certifications

- **Best Innovation Award** – Awarded for making robust obstacle avoidance system for E-HPV by ASME India. **Badges**
- **National Finalist** – Positioned 4th nationally against 1500+ teams in Mathworks Minidrone Competition. **Certificate**
- **Machine Learning Specialization** – by Stanford University and DeepLearning.AI. **Certificate**

Technical Skills

- **Languages:** C, C++, Python, Bash Scripting
- **Frameworks:** Tensorflow, Pytorch, Scikit Learn, Numpy, Pandas, Matplotlib
- **Softwares:** MATLAB, Anaconda - Jupyter Notebook, Google Colab, LTSpice

Extracurricular Activities

Axiom

June 2024 – June 2025

Chief Event Coordinator & Treasurer

NITR Maths Club

- Organised *Synapse-2023*, the annual mathematics competition, where 450 students participated from Rourkela City.